Running of the Spectral Index for Ornstein-Zernike Propagators: An Exact Algebraic Identity and Its Implications

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Abstract

We establish an exact algebraic identity for the Ornstein-Zernike propagator $P(K) = T/(JK^2 + m^2)$ with K-independent parameters: the running of the spectral index $\alpha_s \equiv dn_s/d\ln K$ is a quadratic function of the spectral index itself,

$$\alpha_s = n_s^2 - 1.$$

The derivation is four lines of algebra from the propagator form; the result is parameter-free and holds in any physical context where an O-Z correlator determines a power-law scaling index. We generalize to arbitrary power-law dispersion $P(K) = T/(JK^a + m^a)$, obtaining $\alpha_s = -(a-1+n_s)(1-n_s)$, of which $a = 2$ is a special case. Five independent arguments establish that within the Josephson phase sector of a compact lattice with spontaneously broken $U(1)$, corrections to the identity are parametrically suppressed: multi-pole structure ($|\delta\alpha_s| = 5.8 \times 10^{-9}$), running mass (bounded by $\gamma < 1 - n_s$), zero-mode protection, RPA vertex corrections ($|\delta\alpha_s| = 1.1 \times 10^{-5}$), and Goldstone dispersion.

If the CMB power spectrum is identified with an O-Z propagator at lattice scale $K_{\text{pivot}} \sim M_{\text{KK}}$, then at the Planck 2018 extended-model value $n_s = 0.9625 \pm 0.0048$, the identity predicts $\alpha_s = -0.0734 \pm 0.0092$. The Planck measurement $\alpha_s = -0.0045 \pm 0.0067$ disagrees at 5.0–6.0σ (depending on the treatment of the Planck n_s-α_s correlation). However, this tension is conditional on the scale mapping: at the physical CMB pivot $k_* = 0.05$ Mpc^{-1}, the naive mapping gives $K_{\text{pivot}} \sim 10^{-57}\, M_{\text{KK}}$, at which $n_s \to 1$ and $\alpha_s \to 0$, consistent with any measurement. The lattice-scale identification has not been derived from first principles. We present both the mathematical content (which is exact) and the physical implications (which are conditional), and identify the spectral action correlator as a candidate for identity-breaking with effective dispersion $\alpha_{\text{eff}} = 1.21 \neq 2$.

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1 Introduction

The scalar spectral index n_s and its running $\alpha_s \equiv dn_s/d\ln k$ are among the most precisely measured parameters in cosmology. Planck 2018 reports $n_s = 0.9649 \pm 0.0042$ and $\alpha_s = -0.0045 \pm 0.0067$, the latter consistent with zero Planck Collaboration [2020]. In standard slow-roll inflation, α_s is second order in slow-roll parameters and generically of order $(1 - n_s)^2 \sim 10^{-3}$, well below current sensitivity Liddle and Lyth [2000].

In standard treatments, n_s and α_s are independent parameters determined by the inflaton potential and its derivatives. No algebraic relation of the form $\alpha_s = f(n_s)$ exists within single-field slow-roll models. The purpose of this paper is to exhibit a class of propagators for which such a relation *does* hold as an exact algebraic identity, and to confront the resulting prediction with data.

The propagators in question are of Ornstein-Zernike form,

$$P(K) = \frac{T}{J\,K^2 + m^2}, \tag{1}$$

arising naturally as the two-point correlation function of the Goldstone phase field on a compact Josephson lattice with spontaneously broken $U(1)$ symmetry. Such lattices appear in the phonon-exflation framework MemeTheory and Claude Opus 4.6 [2026], where the internal geometry of a Kaluza-Klein compactification on $SU(3)$ undergoes a Jensen deformation, and the resulting BCS condensate in the Dirac spectrum spontaneously breaks a $U(1)$ symmetry generated by a Kosmann vector field K_7.

The central mathematical result is:

Proposition 1. *For any propagator of the form* (1) *with K-independent T, J, and m, the spectral index $n_s(K) \equiv d\ln P/d\ln K + 1$ and its running $\alpha_s \equiv dn_s/d\ln K$ satisfy*

$$\alpha_s = n_s^2 - 1. \tag{2}$$

This identity is parameter-free: T, J, and m drop out entirely. Once n_s is specified by observation, α_s is determined with no remaining freedom. The identity trivially generalizes to $P(K) = T/(JK^a + m^a)$, yielding $\alpha_s = -(a - 1 + n_s)(1 - n_s)$; the $a = 2$ case is one instance. The identity is therefore a kinematic property of any power-law propagator, not a dynamical prediction. Its physical content lies in the *conditional* statement: if the CMB power spectrum at the pivot scale is described by an O-Z correlator, then α_s is fixed by n_s alone.

At the Planck extended-model value $n_s = 0.9625$, the identity gives $\alpha_s = -0.073$, a factor of 16 larger in magnitude than the Planck central value. Whether this constitutes a tension depends on the scale mapping between the CMB pivot and the internal lattice, as discussed in Section 5.

The paper is organized as follows. Section 2 defines the K^2 propagator on a compact Josephson lattice and establishes notation. Section 3 presents the derivation of the identity and its generalization to K^a dispersion. Section 4 establishes the robustness of the identity against corrections within the O-Z sector. Section 5 discusses the scale dependence of the prediction and confronts it with Planck data. Section 6 discusses the regime of validity, identifies mechanisms that can break the identity, and assesses the physical status of the result.

2 Setup: K^2 Propagators on Compact Lattices

2.1 The Josephson lattice

Consider a periodic lattice of N_{cell} identical sites, each carrying a complex order parameter $\Delta_i = |\Delta_i|\, e^{i\phi_i}$ describing a BCS condensate. The sites are coupled by nearest-neighbor Josephson tunneling with coupling constants J_{ij}. In the framework under consideration, the lattice arises from the tessellation of an $SU(3)$ group manifold into $N_{\text{cell}} = 32$ cells under a $\mathbb{Z}_3$ partition, with two distinct coupling constants: $J_{xy} = J_{C^2} = 0.933\, M_{\text{KK}}$ within the C^2-planes and $J_z = J_{su(2)} = 0.059\, M_{\text{KK}}$ between planes, giving an anisotropy ratio $J_{xy}/J_z = 15.8$ Project computation [2026a].

When the condensate amplitude $|\Delta_i|$ is approximately uniform across cells (the Josephson regime), the low-energy dynamics is governed by the phase fluctuations $\delta\phi_i = \phi_i - \bar{\phi}$. The effective Hamiltonian is

$$H_{\text{eff}} = \sum_{\langle ij \rangle} J_{ij}\big(1 - \cos(\phi_i - \phi_j)\big) + \frac{m_G^2}{2} \sum_i \phi_i^2, \tag{3}$$

where the mass term m_G arises from explicit symmetry breaking (in the framework, from the Leggett dipolar coupling between sectors at $m_G = \omega_{L1} = 0.070\, M_{\text{KK}}$). For small phase fluctuations, this reduces to a massive lattice Klein-Gordon theory with dispersion

$$\omega^2(\mathbf{K}) = \sum_\mu 4J_\mu \sin^2\!\Big(\frac{K_\mu}{2}\Big) + m_G^2, \tag{4}$$

which in the continuum limit $K_\mu \ll \pi$ becomes $\omega^2 \approx J_{\text{eff}}\, K^2 + m_G^2$ with $J_{\text{eff}} = (2J_{xy} + J_z)/3$.

2.2 The Ornstein-Zernike propagator

The static two-point correlator of the phase field in thermal equilibrium at temperature T is

$$P(K) \equiv \langle |\phi(\mathbf{K})|^2 \rangle = \frac{T}{J_{\text{eff}}\, K^2 + m^2}, \tag{5}$$

where we define the effective mass parameter m^2 to absorb the bare mass and any K-independent self-energy contributions. This is the Ornstein-Zernike form Ornstein and Zernike [1914], the universal correlator for a massive scalar field in equilibrium. In the framework, $T = T_{\text{acoustic}} = 0.112\, M_{\text{KK}}$.

If this correlator is identified with the primordial power spectrum $\mathcal{P}(k) \propto P(K)$ via a mapping $k \to K$, then the spectral index and its running are defined by

$$n_s - 1 \equiv \frac{d\ln P}{d\ln K}, \qquad \alpha_s \equiv \frac{dn_s}{d\ln K} = \frac{d^2 \ln P}{(d\ln K)^2}. \tag{6}$$

2.3 The pivot scale and mass parameter

The Planck pivot scale $k_* = 0.05\,\text{Mpc}^{-1}$ maps to some wavenumber K_{pivot} on the lattice. The mapping depends on the expansion history and is a load-bearing assumption of the

framework (see Section 6). For the present analysis, we note that the spectral index at K_{pivot} determines the mass parameter through

$$n_s = 1 - \frac{2}{1+y}, \qquad y \equiv \frac{m^2}{J_{\text{eff}}\, K_{\text{pivot}}^2}. \tag{7}$$

At $n_s = 0.965$, this gives $y = (1+n_s)/(1-n_s) = 56.14$, requiring $m^2 = 56.14\, J_{\text{eff}}\, K_{\text{pivot}}^2$. For the framework parameters, $m_* = 11.87\, M_{\text{KK}}$, which exceeds the physical Leggett mass $m_G = 0.070\, M_{\text{KK}}$ by a factor of 170. This mass problem is entangled with the α_s prediction and is discussed in Section 6.

3 Derivation of the Structural Relation

Proof of Proposition 1. From (5),

$$\ln P(K) = \ln T - \ln\bigl(JK^2 + m^2\bigr). \tag{8}$$

Define the dimensionless ratio

$$y \equiv \frac{m^2}{JK^2}. \tag{9}$$

Then

$$n_s - 1 = \frac{d\ln P}{d\ln K} = -\frac{2JK^2}{JK^2 + m^2} = -\frac{2}{1+y}. \tag{10}$$

Solving for y:

$$y = \frac{1+n_s}{1-n_s}. \tag{11}$$

Since $y = m^2/(JK^2)$ and m, J are K-independent by assumption, $y \propto K^{-2}$, so

$$\frac{dy}{d\ln K} = -2y. \tag{12}$$

The running is

$$\begin{aligned} \alpha_s = \frac{dn_s}{d\ln K} &= \frac{d}{d\ln K}\left(1 - \frac{2}{1+y}\right) = \frac{2}{(1+y)^2}\,\frac{dy}{d\ln K} \\ &= \frac{2}{(1+y)^2}\cdot(-2y) = -\frac{4y}{(1+y)^2}. \end{aligned} \tag{13}$$

Substituting $1 + y = 2/(1-n_s)$ from (11):

$$\begin{aligned} \alpha_s &= -4\cdot\frac{1+n_s}{1-n_s}\cdot\frac{(1-n_s)^2}{4} \\ &= -(1+n_s)(1-n_s) = -(1-n_s^2). \end{aligned} \tag{14}$$

Therefore

$$\boxed{\alpha_s = -(1-n_s^2) = n_s^2 - 1.} \tag{15}$$

□

Corollary 2 (Generalization to K^a dispersion)**.** *For a propagator with power-law dispersion $P(K) = T/(J\,K^a + m^a)$ with $a > 0$ and K-independent parameters, define $n_s - 1 = d\ln P/d\ln K$. Then*

$$\alpha_s = -(a - 1 + n_s)(1 - n_s). \tag{16}$$

Proof. With $y \equiv m^a/(JK^a)$, one has $n_s - 1 = -a/(1+y)$ and $dy/d\ln K = -a\,y$. Then $\alpha_s = a^2 y/(1+y)^2 \cdot (-1) = -(n_s - 1)(a + n_s - 1) = -(a - 1 + n_s)(1 - n_s)$. □

Remark 3. For $a = 2$, the corollary reduces to $\alpha_s = -(1+n_s)(1-n_s) = n_s^2 - 1$. The identity is therefore not specific to K^2 dispersion — it is a property of *any* single-pole propagator with power-law dispersion. The $a = 2$ case is distinguished physically (Goldstone theorem) but not algebraically. This underscores that the identity is a kinematic consequence of the propagator form, not a dynamical prediction.

Remark 4. The identity holds for *any* positive values of T, J, m, and K. These four parameters determine y and hence n_s at a given scale, but α_s depends only on n_s regardless of which particular (T, J, m, K) produced that n_s. This is because the Ornstein-Zernike propagator is a one-parameter family when expressed in terms of $y = m^2/(JK^2)$, and both n_s and α_s are functions of y alone.

Remark 5. The relation $\alpha_s = n_s^2 - 1$ also follows from the observation that $P(K) \propto K^{n_s - 1}$ locally, with $n_s(K)$ a running exponent. For the O-Z form, $n_s - 1 = -2/(1+y)$ is a Möbius transformation of $y = m^2/(JK^2)$. The composition of a Möbius transformation with the logarithmic derivative yields the quadratic relation. This is an algebraic inevitability, not a dynamical coincidence.

3.1 Explicit assumptions

The derivation requires exactly three assumptions:

A1. The propagator has the Ornstein-Zernike form $P(K) = T/(JK^2 + m^2)$.

A2. The parameters T, J, and m are independent of K.

A3. The spectral index is defined as $n_s - 1 = d\ln P/d\ln K$.

Any violation of **A1** (e.g., multi-pole structure, non-K^2 dispersion) or **A2** (e.g., running mass $m(K)$, running stiffness $J(K)$, or running temperature $T(K)$) can in principle break the identity. Section 4 establishes that within the Josephson phase sector of the compact lattice framework, all five physically motivated mechanisms for violating **A1** or **A2** produce corrections that are parametrically suppressed.

4 Robustness of the Identity within the O-Z Sector

The identity (15) holds exactly for the bare propagator. Physical corrections could modify it if they introduce K-dependence into T, J, m, or if they change the functional form away from K^{-2}. We examine five independent mechanisms and show that each produces corrections to α_s that are negligible compared to the tension with Planck.

Two of these arguments are *structural*: they hold for any O-Z propagator regardless of the underlying physical system (Proof 4.2, the running mass bound; Proof 4.5, the Goldstone theorem). The remaining three are *framework-specific*: they depend on the numerical values of the Josephson lattice parameters (Proofs 4.1, 4.3, 4.4). Together, the five arguments establish that corrections *within the O-Z sector* cannot break the identity. They do *not* establish that the CMB power spectrum belongs to the O-Z sector.

4.1 Proof 1: Three-pole Leggett propagator

The BCS condensate on the Jensen-deformed $SU(3)$ has three sectors (B_1, B_2, B_3) with distinct gaps, coupled by inter-sector Josephson terms. The full propagator is therefore a sum of three poles:

$$P_{3-\text{pole}}(K) = \sum_{i=1}^{3} \frac{w_i}{JK^2 + \lambda_i}, \tag{17}$$

where $\lambda_i = m_{\text{base}}^2 + \sigma_i$ are the eigenvalues of the 3×3 mass matrix (σ_i being the Josephson splittings) and w_i are the spectral weights. This is a violation of assumption **A1**: a sum of Ornstein-Zernike propagators with distinct masses is not itself Ornstein-Zernike.

However, the pole splitting is parametrically suppressed by the mass hierarchy. The physical values are Project computation [2026d]:

$$m_{\text{base}}^2 = 140.5\, M_{\text{KK}}^2, \qquad \sigma_{\max} = 0.072\, M_{\text{KK}}^2, \qquad \frac{\sigma_{\max}}{m_{\text{base}}^2} = 5.1 \times 10^{-4}. \tag{18}$$

The poles are 99.95% degenerate. By the equal-stiffness theorem (all three sectors share the same spatial Josephson coupling from the underlying lattice geometry), the weights satisfy $w_1 = w_2 = w_3 = 1/3$ to leading order, and the three-pole propagator decomposes as

$$P_{3-\text{pole}}(K) = \frac{1}{3} \sum_{i=1}^{3} P_{\text{OZ}}(K;\, \lambda_i). \tag{19}$$

The identity $\alpha_s = n_s^2 - 1$ holds *individually* for each term. The correction from the multi-pole sum is proportional to $dw_i/d\ln K \cdot (n_{s,i} - 1)$, which is of order $\sigma_{\max}/m_{\text{base}}^2 \sim 5 \times 10^{-4}$. Numerical evaluation gives $|\delta\alpha_s| = 5.8 \times 10^{-9}$, which is nine orders of magnitude below the required shift of $\sim$0.06 to reach Planck compatibility.

A scan over Josephson coupling strengths from $0\times$ to $4000\times$ the physical value confirms that α_s shifts by less than 0.3% even at unphysically large pole splittings (238% splitting). The identity is protected by the mass hierarchy: the $n_s = 0.965$ constraint forces $m_{\text{base}}^2 \gg \sigma_{\max}$, making the poles effectively degenerate.

4.2 Proof 2: Running mass bound

Suppose the mass acquires a power-law running from radiative corrections:

$$m^2(K) = m_0^2 \left(\frac{K}{K_0}\right)^{\gamma}, \tag{20}$$

where γ is the anomalous dimension. This violates assumption **A2**. The propagator becomes $P(K) = T/(JK^2 + m_0^2(K/K_0)^\gamma)$, and the spectral index generalizes to

$$n_s - 1 = -\frac{2 + \gamma\, u}{1 + u}, \qquad u \equiv \frac{m^2(K)}{JK^2}. \tag{21}$$

For $u > 0$ (a physical, non-tachyonic propagator) and $n_s < 1$ (a red-tilted spectrum), this requires

$$\boxed{\gamma < 1 - n_s.} \tag{22}$$

At $n_s = 0.965$: $\gamma < 0.035$.

The correction to the running is $\delta\alpha_s = \gamma(2 - \gamma)\, u/(1 + u)$. At the structural maximum $\gamma \to 0.035$, the correction $\delta\alpha_s \to 0.069$, which would shift α_s from -0.069 to approximately zero. But this limit requires $u \to \infty$, meaning $JK^2/m^2 \to 0$: the kinetic term vanishes, and the propagator degenerates to $P \sim K^{-\gamma}$, which is no longer Ornstein-Zernike.

The physical 1-loop computation on the 32-cell lattice with coupling $\lambda = V(B_2, B_2) = 0.1557$ (the BCS Casimir interaction) gives the sunset self-energy anomalous dimension $\gamma = -6.76 \times 10^{-4}$. This is:

- $52\times$ below the structural bound,
- of the *wrong sign* (mass decreasing at higher K, worsening the tension),
- producing $|\delta\alpha_s| = 1.3 \times 10^{-3}$, two orders of magnitude below the required shift.

The bound $\gamma < 1 - n_s$ is algebraic and holds at all loop orders. Any single-pole propagator consistent with $n_s = 0.965$ has its anomalous dimension constrained to a window $49\times$ too narrow to resolve the tension.

4.3 Proof 3: Zero-mode protection

The Goldstone mode on the compact lattice is the $n = 0$ Kaluza-Klein mode of the phase field on the internal torus T^2. Its wavefunction is *constant* across the fiber. Scattering off the condensate texture $V(\mathbf{x}) = (|\Delta(\mathbf{x})|^2 - \langle|\Delta|^2\rangle)/\langle|\Delta|^2\rangle$ on T^2 produces an eikonal self-energy $\Sigma(K)$.

The zero-mode couples only to the spatial average of the texture potential:

$$\langle n = 0|V|n = 0\rangle = \frac{1}{\mathrm{Vol}(T^2)} \int_{T^2} V(\mathbf{x})\, d^2x = 0, \tag{23}$$

where the last equality holds by construction: $\langle V\rangle = 0$ (verified numerically at 4.5×10^{-17}). Therefore $\Sigma(K_{\mathrm{pivot}}) = 0$ identically, and the self-energy introduces no K-dependence at the Goldstone scale.

This result extends to the full Born series for $\mathrm{Re}\,\Sigma$. The Ward identity associated with the broken $U(1)_7$ forces $\Sigma(0, 0) = 0$: the zero-mode Goldstone is transparent to the texture at all orders in the Born expansion Project computation [2026f].

The physical reason is a scale hierarchy: the Goldstone wavelength $\lambda_G = 2\pi/K_{\mathrm{pivot}} = 67.8\, M_{\mathrm{KK}}^{-1}$ exceeds the torus circumference $L_{T^2} = 13.2\, M_{\mathrm{KK}}^{-1}$ by a factor of 5.2. The Goldstone

mass shell $K_G = 0.093\, M_{\rm KK}$ sits below the torus fundamental mode $K_{\rm fund} = 0.477\, M_{\rm KK}$. There is zero texture power at the Goldstone scale. Higher KK modes ($n \geq 1$), with wavelengths comparable to the texture scale, see strong scattering (Ioffe-Regel parameter $kl = 0.0095 \ll 1$, Anderson localization with $\xi_{\rm loc} = 0.104\, M_{\rm KK}^{-1}$). But these modes are separated from the Goldstone by the KK mass gap and do not contribute to the low-K propagator.

4.4 Proof 4: RPA suppression

The Random Phase Approximation (RPA) resums the pair susceptibility bubble chain into a dressed propagator:

$$D_{\rm RPA}(K) = \frac{D_0(K)}{1 - g^2\, \chi_0(K)\, D_0(K)}, \tag{24}$$

where $D_0(K) = T/(JK^2 + m^2)$ is the bare propagator, $g = V(B_2, B_2) = 0.1557$ is the pair coupling, and $\chi_0(K)$ is the static pair susceptibility computed from the BdG quasiparticle spectrum:

$$\chi_0(K) = \sum_n \rho_n \frac{F_n^2}{2E_{\rm qp}(n) + J_{\rm pair}\, \epsilon(K)\, w_n}, \tag{25}$$

with $F_n = 2u_n v_n$ the BCS coherence factor and $\epsilon(K) = 2(1 - \cos K l_{\rm cell})$ the tight-binding dispersion.

If $\chi_0(K)$ has significant K-dependence, it would modify the slope of $D_{\rm RPA}$ relative to D_0 and break the identity. The computation reveals three independent suppression mechanisms:

1. **Flat susceptibility.** $\chi_0(K)$ varies by only 0.30% across the fitting window $[0.85, 1.15]\, K_{\rm pivot}$. The effective power-law index $d\ln\chi_0/d\ln K = -0.0098 \approx 0$. The pair coherence length $\xi_{\rm BCS} = 0.81\, M_{\rm KK}^{-1}$ spans 5.3 cells, but the pair dispersion bandwidth $4J_{\rm pair} = 0.461\, M_{\rm KK}$ is only 2.7% of the pair gap $2E_{\rm qp,min} = 1.70\, M_{\rm KK}$. The pair is *stiff*: it cannot respond to the slow spatial modulation at $K_{\rm pivot}$.

2. K^2 **renormalization only.** The leading K-dependence of χ_0 is proportional to K^2 (from the tight-binding dispersion), which renormalizes $J_{\rm eff}$ but preserves the $K^2 + m^2$ form. Lattice corrections are $O((Kl_{\rm cell})^2/12) \sim 7.5 \times 10^{-3}$.

3. **Mass hierarchy suppression.** Even with $g^2\chi_0 = O(1)$ (the physical value is 0.51, comparable to nuclear effective charge couplings), the propagator $D_0(K_{\rm pivot}) = 7.8 \times 10^{-4}$ is suppressed by the mass hierarchy $m^2/(JK^2) = 56$. The RPA self-energy correction $\Pi \cdot D_0 = 4.0 \times 10^{-4}$, giving $D_{\rm RPA}/D_0 - 1 = 0.04\%$.

The net RPA correction to α_s is $+1.1 \times 10^{-5}$, nine orders of magnitude below the required shift. Scanning g from 0 to 2.0 (13× the Casimir value) shows $|\delta\alpha_s| < 2 \times 10^{-4}$: the identity is protected by the mass hierarchy regardless of coupling strength.

4.5 Proof 5: Goldstone theorem

The Goldstone theorem guarantees that for any system with a spontaneously broken continuous symmetry, the low-energy excitations are massless Nambu-Goldstone modes with dispersion $\omega^2 \propto K^2$ in the long-wavelength limit. This is a structural consequence of the broken symmetry, independent of microscopic details.

On the 32-cell lattice, three models were constructed to test whether lattice disorder can induce anomalous dispersion $\omega^2 \propto K^\alpha$ with $\alpha \neq 2$:

1. **Isotropic**: $J = J_{\text{eff}} = 0.641\, M_{\text{KK}}$ for all bonds.
2. **Anisotropic**: $J_{xy} = 0.933$, $J_z = 0.059\, M_{\text{KK}}$ (ratio 15.8).
3. **$\mathbb{Z}_3$-disordered**: Same as (2), but bonds crossing $\mathbb{Z}_3$ domain walls weakened to $J_{xy}/4 = 0.233\, M_{\text{KK}}$ (75% of in-plane bonds are cross-domain).

Exact diagonalization of the 32×32 lattice Laplacians in all three cases gives the effective dispersion exponent $\alpha_{\text{eff}} = 2.0$ at K_{pivot}. The $\mathbb{Z}_3$ disorder softens the effective stiffness (bandwidth reduced by factor 0.57) and creates low-energy states, but the functional form remains K^2. The propagator ratio $P_{\mathbb{Z}_3}(K)/P_{\text{iso}}(K)$ is flat to 1.7% across all 14 K-shells. The disorder changes the effective mass (renormalization) but not the dispersion (anomalous exponent).

For anomalous dispersion ($\alpha \neq 2$) to arise from lattice disorder, one requires either: (a) random long-range correlated disorder producing Lévy flights, (b) Anderson localization on exponentially many sites, or (c) non-local couplings beyond nearest-neighbor. The $\mathbb{Z}_3$ structure is a periodic superlattice, not random disorder, and satisfies none of these conditions. The lattice correction to K^2 at K_{pivot} is 0.75%.

Remark 6. The five proofs are independent and address different potential violations of assumptions **A1**–**A2**. Proof 1 addresses multi-pole structure. Proof 2 addresses running parameters. Proof 3 addresses self-energy corrections. Proof 4 addresses vertex corrections. Proof 5 addresses the dispersion relation itself. The largest correction among all five is $|\delta\alpha_s| = 1.3 \times 10^{-3}$ (from Proof 2), which is 50× below the required 0.064 shift.

5 Comparison to Observation

5.1 Planck 2018 constraints

The identity predicts a specific α_s for each n_s, so the correct observational comparison uses the Planck 2018 *extended model* in which α_s is a free parameter (not fixed to zero). The Planck collaboration reports for the ΛCDM+α_s model Planck Collaboration [2020]:

$$n_s = 0.9625 \pm 0.0048 \quad (68\%\ \text{CL}), \tag{26}$$

$$\alpha_s = -0.0045 \pm 0.0067 \quad (68\%\ \text{CL}), \tag{27}$$

with correlation $\rho(n_s, \alpha_s) \approx -0.5$. (The base ΛCDM fit, which fixes $\alpha_s = 0$, gives $n_s = 0.9649 \pm 0.0042$; using those values would overstate precision by ignoring the n_s-α_s degeneracy.)

5.2 Prediction from the structural relation

At the extended-model central value $n_s = 0.9625$:

$$\alpha_s^{\text{pred}} = n_s^2 - 1 = (0.9625)^2 - 1 = -0.0734. \tag{28}$$

The uncertainty propagates through $d\alpha_s/dn_s = 2n_s = 1.925$:

$$\sigma(\alpha_s^{\text{pred}}) = 2n_s \cdot \sigma(n_s) = 1.925 \times 0.0048 = 0.0092. \tag{29}$$

The full posterior, computed via Monte Carlo with 10^4 samples drawing n_s from a Gaussian prior, gives Project computation [2026b]:

$$\alpha_s^{\text{pred}} = -0.0734 \pm 0.0092 \quad (68\% \text{ CI}). \tag{30}$$

The 95% CI excludes $\alpha_s = 0$ entirely.

5.3 Scale dependence of the prediction

The identity $\alpha_s = n_s^2 - 1$ holds at *every* scale K. However, the *predicted value* of α_s depends on which internal wavenumber K_{pivot} the CMB pivot scale $k_* = 0.05$ Mpc^{-1} maps to. This mapping is the load-bearing assumption that separates the mathematical identity from its physical consequences.

At the physical CMB scale: Taking $M_{\text{KK}} \sim M_{\text{Pl}} \sim 10^{19}$ GeV and $k_* = 0.05$ Mpc$^{-1} \sim 10^{-38}$ GeV, the naive mapping gives

$$K_{\text{pivot}} \sim \frac{k_*}{M_{\text{KK}}} \sim 10^{-57}\ M_{\text{KK}}. \tag{31}$$

At this K, the dimensionless ratio $y = m^2/(JK_{\text{pivot}}^2)$ is astronomically large ($y \sim 10^{114}$), giving $n_s = 1 - 2/(1+y) = 1 - O(10^{-114})$ and $\alpha_s = n_s^2 - 1 = -O(10^{-114})$. This is consistent with *any* measurement: the O-Z propagator at the physical CMB scale is indistinguishable from a flat spectrum.

At the lattice scale: The computation in this paper uses $K_{\text{pivot}} = 2\pi/N_{\text{cell}}^{1/3} \sim 2.0\ M_{\text{KK}}$, the fundamental mode of the 32-cell lattice. This identification produces $n_s = 0.965$ with $m_* = 11.87\ M_{\text{KK}}$, yielding the $\alpha_s = -0.073$ prediction.

Status: The lattice-scale identification $K_{\text{pivot}} \sim M_{\text{KK}}$ has *not* been derived from first principles. It requires either (a) a mechanism that imprints the internal lattice correlations onto the CMB at a specific epoch, or (b) a non-trivial scale mapping $k \to K(k)$ that compresses 57 orders of magnitude. Neither has been constructed within the framework. The 5.0–6.0σ tension reported below is therefore *conditional* on this identification.

5.4 Tension with observation

Treating the predicted and observed α_s as independent (a conservative choice; the Planck n_s-α_s anti-correlation $\rho \approx -0.5$ would modify the combined uncertainty):

$$\frac{|\alpha_s^{\text{pred}} - \alpha_s^{\text{obs}}|}{\sqrt{\sigma_{\text{pred}}^2 + \sigma_{\text{obs}}^2}} = \frac{|-0.073 - (-0.005)|}{\sqrt{0.0092^2 + 0.0067^2}} = \frac{0.069}{0.0114} = 6.0\sigma. \tag{32}$$

Including the n_s-α_s correlation from the Planck chain (the prediction $\alpha_s^{\rm pred} = n_s^2 - 1$ inherits the Planck n_s measurement, which is correlated with $\alpha_s^{\rm obs}$), the combined variance becomes $\sigma_\Delta^2 = \sigma_{\rm pred}^2 + \sigma_{\rm obs}^2 - 2 \cdot 2n_s \cdot \rho \cdot \sigma_{n_s} \cdot \sigma_{\alpha_s}^{\rm obs}$, giving $\sigma_\Delta = 0.014$ and a tension of 5.0σ. The tension is 5.0–6.0σ depending on the treatment of correlations.

Critical caveat: This tension assumes $K_{\rm pivot} \sim M_{\rm KK}$. See Section 5.3 for why this assumption is load-bearing and what happens without it.

We stress three features:

1. **The prediction has zero theoretical uncertainty.** The only source of $\alpha_s^{\rm pred}$ uncertainty is the Planck n_s measurement error. The Josephson couplings J_{ij}, the pivot wavenumber $K_{\rm pivot}$, and the mass m_* all cancel identically in the α_s formula. Monte Carlo with 50% log-normal priors on J_{xy}, 80% on J_z, and 30% on $K_{\rm pivot}$ gives Pearson $r < 0.015$ with α_s for each; all R^2 contributions are 0.0%. The n_s-α_s correlation is $r = +1.000$.

2. **The lattice does not help.** On a finite lattice with $N_{\rm cell} = 32$ and lattice dispersion $\sin^2(K/2)$ replacing K^2, the numerical running changes from -0.073 (continuum) to -0.038 (lattice $N = 32$), reducing the tension. However, this lattice correction is an artifact of evaluating the derivative on 14 discrete K-shells; in the angular-averaged analysis with proper continuum extrapolation, the continuum value is correct.

3. **CMB-S4 is decisive.** With $\sigma(\alpha_s) \sim 0.003$, CMB-S4 would detect $\alpha_s = -0.073$ at $\sim 8\sigma$ if the lattice-scale identification holds. If CMB-S4 measures $\alpha_s = 0.000 \pm 0.003$, the O-Z identification at lattice scale is excluded at $>20\sigma$.

5.5 Summary of observational status

Observable	Identity prediction	Planck 2018 (extended)	Tension
n_s	input	0.9625 ± 0.0048	—
α_s	-0.073 ± 0.009	-0.005 ± 0.007	5.0–6.0σ
r	0	< 0.06	consistent

Table 1: Comparison of identity predictions to Planck 2018 extended-model (ΛCDM+α_s) measurements. The spectral index n_s is an input (one free parameter m), not a prediction. The running α_s is a zero-parameter consequence of the identity, conditional on the lattice-scale mapping $K_{\rm pivot} \sim M_{\rm KK}$. The tension range reflects the treatment of the Planck n_s-α_s correlation.

6 Discussion

6.1 Regime of validity

The structural relation $\alpha_s = n_s^2 - 1$ holds exactly under assumptions **A1**–**A3** of Section 3. Its domain of validity is:

- Any scalar two-point correlator of Ornstein-Zernike form with K-independent parameters.
- Any lattice geometry, coupling constants, temperature, and mass. These parameters set the scale at which a given n_s is achieved, but the relation between n_s and α_s is universal within the class.
- The continuum limit $K \ll \pi/a$ (lattice spacing a). At finite lattice spacing, corrections of order $(Ka)^2$ enter; see Section 5.

The $a = 2$ relation does *not* hold for:

- Propagators with more than one pole at widely separated masses (pole spread $\gtrsim O(1)$).
- Propagators with dispersion $\omega^2 \propto K^\alpha$, $\alpha \neq 2$ (though the generalized identity from Corollary 2 applies).
- Any correlator with K-dependent $T(K)$, $J(K)$, or $m(K)$ having anomalous dimension $\gamma \gtrsim 1 - n_s$.

6.2 What breaks the identity: the spectral action correlator

A correlator outside the Josephson phase sector that *does* break the identity has been identified within the framework Project computation [2026c]. The spectral action two-point function

$$\chi_{\mathrm{SA}}(K) = \sum_{(p,q)} \frac{W_{(p,q)}}{K^2 + C_2(p,q)} \tag{33}$$

is a sum of Ornstein-Zernike propagators with masses set by the quadratic Casimir eigenvalues $C_2(p,q) = (p^2 + q^2 + pq + 3p + 3q)/3$ of $SU(3)$ representations. The Casimir eigenvalues range from 1.33 to 9.33, a 110% spread compared to the 0.051% pole splitting of the Josephson propagator. This large spread produces an effective dispersion exponent $\alpha_{\mathrm{eff}} = 1.21 \neq 2$ at K_{pivot}, breaking the identity.

However, the spectral action correlator alone gives $n_s \approx 0.21$ (far too red). A physical power spectrum would require a mixture of the Josephson and spectral action correlators:

$$P_{\mathrm{CMB}}(K) = a\, P_{\mathrm{Goldstone}}(K) + b\, \chi_{\mathrm{SA}}(K), \tag{34}$$

with mixing coefficients a, b determined by the physical mechanism that imprints perturbations during the transit. For $\beta \equiv b/(a+b) > 0.9$, the convex combination theorem allows $n_s = 0.965$ at $K < K_* = m_G/\sqrt{J} = 0.087\, M_{\mathrm{KK}}$ with α_s in the Planck window Project computation [2026e]. The critical question — which correlator the CMB measures — reduces to the coupling between the spectral action and Josephson sectors, which has not been computed from first principles.

6.3 The mass problem

The identity relates α_s to n_s *at a given scale.* Achieving $n_s = 0.965$ requires $m_* = 11.87\, M_{\mathrm{KK}}$, which exceeds the physical Leggett mass $m_G = 0.070\, M_{\mathrm{KK}}$ by a factor of 170. No mechanism within the framework produces this mass.

The mass problem is logically *entangled* with the α_s prediction: the $n_s = 0.965$ value is imposed as a boundary condition (by fitting m_* to match the Planck spectral index), not derived from first principles. Any mechanism that modifies the propagator at the pivot scale to resolve the mass problem will also modify the running. Conversely, the five proofs of Section 4 demonstrate that modifications *within* the Josephson phase sector are insufficient for either problem — they cannot bring m_* down to m_G, nor can they shift α_s to the Planck window.

6.4 The scale mapping problem

The scale mapping between the CMB pivot and the internal lattice is discussed quantitatively in Section 5.3. To summarize: the physical CMB pivot maps to $K_{\mathrm{pivot}} \sim 10^{-57}\, M_{\mathrm{KK}}$, at which $n_s \to 1$ and no tension exists. The lattice computation uses $K_{\mathrm{pivot}} \sim 2\, M_{\mathrm{KK}}$, which gives $n_s = 0.965$ and the reported tension. No first-principles derivation of the latter identification exists. Constructing such a mapping — or proving that none exists — is the decisive open problem for the physical interpretation of the identity.

6.5 Status of the result

The result has three distinct levels of content, with decreasing certainty:

1. **Mathematical identity** (exact): the relation $\alpha_s = n_s^2 - 1$ for K^2 propagators, and its generalization $\alpha_s = -(a - 1 + n_s)(1 - n_s)$ for K^a propagators, are algebraic identities. They hold for any O-Z correlator with K-independent parameters, independent of physical interpretation. Applications may exist in condensed matter or statistical mechanics wherever O-Z correlators determine power-law indices.

2. **Conditional prediction** (depends on scale mapping): *if* the CMB power spectrum is identified with an O-Z propagator at lattice scale $K_{\mathrm{pivot}} \sim M_{\mathrm{KK}}$, then $\alpha_s = -0.073$ at $n_s = 0.9625$, in 5.0–6.0σ tension with Planck 2018. This constitutes a falsification of the *pure Josephson phase sector at lattice scale* as the origin of CMB fluctuations. At the physical CMB scale ($K_{\mathrm{pivot}} \sim 10^{-57}\, M_{\mathrm{KK}}$), no tension exists.

3. **Constraint on viable correlators**: the identity restricts the space of viable CMB correlators to those that deviate from K^2 dispersion (if the O-Z identification at lattice scale is assumed). Within the framework, the spectral action correlator satisfies this requirement. More broadly, any cosmological scenario generating primordial fluctuations from an O-Z correlator predicts $\alpha_s = n_s^2 - 1$, which is in tension with data *at any scale where n_s deviates measurably from 1.*

6.6 Absence of prior art

A literature search across standard references on CMB power spectra, inflationary cosmology, statistical mechanics, and Ornstein-Zernike theory found no prior statement of the relation $\alpha_s = n_s^2 - 1$. The derivation is elementary (four lines of algebra from the propagator form). Its absence from the literature may reflect the fact that standard inflationary models generate power spectra from slow-roll dynamics rather than equilibrium correlators, and the Ornstein-Zernike form with constant parameters is not the standard starting point for CMB analyses. In condensed matter, the running of the spectral index is not a commonly studied quantity.

6.7 Comparison to slow-roll inflation

The O-Z relation and slow-roll inflation make qualitatively *different* predictions for the scaling of α_s with the tilt $\epsilon \equiv 1 - n_s$. Writing $n_s = 1 - \epsilon$:

$$\alpha_s^{\text{O-Z}} = (1-\epsilon)^2 - 1 = -2\epsilon + \epsilon^2 \approx -2\epsilon, \tag{35}$$

which is *linear* in the tilt at leading order. In slow-roll inflation, $n_s - 1 = -2\epsilon_H - \eta_H$ and $\alpha_s = -2\epsilon_H\eta_H - \xi_H^2$, which is generically *quadratic* in slow-roll parameters: $\alpha_s \sim O(\epsilon^2)$ for $\epsilon_H \sim \eta_H \sim \epsilon$.

The distinction is stark: the O-Z identity gives $|\alpha_s| \sim 2|\epsilon| \sim 0.07$ at $\epsilon = 0.035$, while generic slow-roll gives $|\alpha_s| \sim \epsilon^2/2 \sim 6 \times 10^{-4}$, two orders of magnitude smaller. The O-Z prediction exceeds slow-roll by a factor of $\sim 2/\epsilon \sim 57$. There is no special case of slow-roll parameters that reproduces the O-Z relation: $\eta_H = 0$, $\xi_H = 0$ would give $\alpha_s = 0$, not $\alpha_s = -2\epsilon$. The linear-in-tilt scaling is a signature that distinguishes O-Z correlators from slow-roll spectra.

7 Conclusion

We have established that the Ornstein-Zernike propagator $P(K) = T/(JK^2 + m^2)$ with K-independent parameters satisfies the exact algebraic identity $\alpha_s = n_s^2 - 1$. The derivation is four lines of algebra; the result is parameter-free. The identity generalizes to $\alpha_s = -(a - 1 + n_s)(1 - n_s)$ for K^a dispersion, of which $a = 2$ is a special case.

Five independent arguments — two structural (running mass bound, Goldstone theorem) and three framework-specific (Leggett pole degeneracy, zero-mode protection, RPA suppression) — establish that within the O-Z sector of a compact lattice with spontaneously broken $U(1)$, the identity cannot be broken. The largest correction is $|\delta\alpha_s| = 1.3 \times 10^{-3}$, compared to the required shift of $\sim$0.07. These arguments do *not* establish that the CMB power spectrum belongs to the O-Z sector.

If the CMB pivot scale maps to the lattice fundamental mode ($K_{\text{pivot}} \sim M_{\text{KK}}$), the identity predicts $\alpha_s = -0.073 \pm 0.009$ at the extended-model $n_s = 0.9625$, in 5.0–6.0σ tension with Planck 2018. This tension is *conditional*: at the physical CMB scale ($K_{\text{pivot}} \sim 10^{-57}\, M_{\text{KK}}$), $n_s \to 1$ and $\alpha_s \to 0$, consistent with any measurement. The lattice-scale identification has not been derived from first principles. Achieving $n_s = 0.965$ at the lattice scale requires $m_* = 11.87\, M_{\text{KK}}$, a factor of 170 above the physical Leggett mass — a mass problem that is logically entangled with the α_s prediction.

The tension, where it exists, can be resolved by correlators with effective dispersion $\omega^2 \propto K^\alpha$, $\alpha \neq 2$. Within the framework, the spectral action correlator, built from Casimir eigenvalues spanning a factor of 7 in mass, has $\alpha_{\text{eff}} = 1.21$ and breaks the identity. Whether this correlator contributes to the CMB power spectrum remains an open computation.

CMB-S4, with $\sigma(\alpha_s) \sim 0.003$, would be decisive for the lattice-scale identification: detection of $\alpha_s = -0.073$ at $\sim 8\sigma$ if the identification holds, or exclusion at $>20\sigma$ if $\alpha_s = 0$.

The mathematical content of the identity — that the Möbius transformation $n_s = 1 - 2/(1+y)$ composed with $y \propto K^{-2}$ yields a quadratic relation between n_s and α_s — is independent of any physical framework and holds in any context where Ornstein-Zernike correlators determine power-law scaling indices.

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